

Samuel Gutknecht

Senior Software Engineer

EXPERIENCE

 Microsoft

Senior Software Engineer

Product engineering for customer experience platforms spanning complex enterprise interfaces, rich-text editing, AI-assisted service experiences, and agent workflows. The work combines React and TypeScript front-end architecture with backend, cloud, diagnostics, and reliability engineering.

 Garmin

Software Engineering Intern

Developed avionics interface software for the Garmin G3000 integrated flight deck using C, C++, and Python. Built and validated interactive graphical widgets defined by the ARINC 661 specification, including C and C++ interface work with TinyGL.

SKILLS & FOCUS

Tools change.
Judgment travels.

AI agent development	LLM & prompt engineering	Context engineering
RAG & vector search	Neural networks & machine learning	Agent evaluations
GitHub Copilot	Semantic Kernel	LangChain
TypeScript	React	C#
Python	JavaScript	C++
C	Computer architecture	Microprocessor systems
Analog & digital circuit design	VHDL & programmable logic	Node.js
SQL	Java	Azure
CI/CD	Git	

EDUCATION

University of Missouri

Bachelor of Science

Electrical and Computer Engineering

2017–2021 · GRADUATED CUM LAUDE

MINORS

Computer Science, Mathematics, and Spanish

TECHNICAL FOUNDATION

Theory in computer systems, electronics, embedded computing, and machine learning

RECOGNITION

Dean's List High Honors · Bright Flight Scholarship · MU Excellence Award